

## SMAPhysics.com JEE MAIN PHYSICS DPP: RELATIVE VELOCITY IN 2D

Topic: River-Boat & Rain-Man Kinematics in Two Dimensions | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A river of width  $d$  flows with speed  $u$ . A swimmer can swim with speed  $v$  ( $v > u$ ) in still water. The time taken to cross the river along the shortest path (zero drift) is:
- $\frac{d}{v}$
  - $\frac{d}{\sqrt{v^2 - u^2}}$
  - $\frac{d}{\sqrt{v^2 + u^2}}$
  - $\frac{d}{v - u}$
- Q2.** Rain is falling vertically downwards with speed 30 m/s. A man cycles due east at 10 m/s. The angle with the vertical at which he should hold his umbrella to protect himself from the rain is:
- $\tan^{-1}(3)$  towards west
  - $\tan^{-1}(3)$  towards east
  - $\tan^{-1}\left(\frac{1}{3}\right)$  towards east
  - $\tan^{-1}\left(\frac{1}{3}\right)$  towards west
- Q3.** A boat crosses a river of width 1 km flowing at 3 km/h. The boat is steered perpendicular to the flow with speed 4 km/h relative to water. The net drift of the boat on reaching the opposite bank is:
- 1.00 km
  - 0.50 km
  - 1.25 km
  - 0.75 km
- Q4.** To cross a river in the shortest possible time, a swimmer should steer:
- Exactly perpendicular to the direction of river flow
  - At an angle  $\sin^{-1}(u/v)$  upstream
  - At an angle  $\sin^{-1}(u/v)$  downstream
  - At  $45^\circ$  upstream
- Q5.** Rain appears to fall vertically to a man walking at 4 km/h. When he doubles his speed to 8 km/h in the same direction, rain appears to fall at  $45^\circ$  to the vertical. The true velocity of rain has magnitude:
- 4 km/h
  - $4\sqrt{2}$  km/h
  - 8 km/h
  - $4\sqrt{3}$  km/h
- Q6.** A river is flowing with velocity  $v_r = 5$  km/h. A boat has speed  $v_{b/r} = 3$  km/h ( $v_{b/r} < v_r$ ). The minimum drift possible for the boat while crossing the river of width  $d = 400$  m is:
- 400 m
  - 600 m
  - 533.3 m
  - 300 m
- Q7.** A man standing on a road has to hold his umbrella at  $30^\circ$  with the vertical to keep rain away. He throws away the umbrella and starts running at 10 km/h in the horizontal direction. He finds that rain drops now hit his head vertically. The actual speed of the rain drops with respect to ground is:
- 10 km/h
  - $10\sqrt{3}$  km/h
  - $\frac{10}{\sqrt{3}}$  km/h
  - 20 km/h
- Q8.** A swimmer wishes to reach a point directly opposite on the other bank of a river flowing at 2 m/s. His swimming speed in still water is 4 m/s. At what angle to the direction of river flow must he direct his strokes?
- $120^\circ$
  - $150^\circ$
  - $60^\circ$
  - $90^\circ$
- Q9.** A boat moves relative to water with velocity  $v$  and river flow velocity is  $u$ . If the time taken by the boat to travel a distance  $x$  downstream and back upstream is  $T_1$ , and the time taken to travel the same distance  $x$  across the river and back along the shortest path is  $T_2$ , the ratio  $T_1/T_2$  is:
- $\sqrt{1 - \frac{u^2}{v^2}}$
  - $\frac{1}{\sqrt{1 - \frac{u^2}{v^2}}}$
  - $1 - \frac{u^2}{v^2}$
  - $\frac{1}{1 - \frac{u^2}{v^2}}$

- Q10.** Rain is falling with a speed of  $12\sqrt{3}$  m/s at an angle of  $30^\circ$  with the vertical towards east. A cyclist rides towards east. If the rain appears to fall vertically to the cyclist, the speed of the cyclist is:
- 12 m/s
  - 18 m/s
  - $6\sqrt{3}$  m/s
  - 6 m/s
- Q11.** A swimmer crosses a river flowing with speed  $u$ . In crossing in minimum time  $T_{\min}$ , he incurs a drift  $x_1$ . In crossing with minimum drift, the time taken is  $T$ . If  $v > u$ , the drift  $x_1$  equals:
- $\frac{vd}{u}$
  - $\frac{ud}{\sqrt{v^2 - u^2}}$
  - $\frac{ud}{v + u}$
  - $\frac{ud}{v}$
- Q12.** A man moves due north at 3 km/h. Rain appears to fall vertically with speed 4 km/h. The actual speed and direction of the rain with respect to ground are:
- 5 km/h,  $\tan^{-1}(3/4)$  south of vertical
  - 5 km/h,  $\tan^{-1}(4/3)$  north of vertical
  - 5 km/h,  $\tan^{-1}(3/4)$  north of vertical
  - 7 km/h,  $45^\circ$  south of vertical
- Q13.** Ship  $A$  is travelling due east at 10 km/h and ship  $B$  is travelling due north at 10 km/h. The velocity of ship  $A$  relative to ship  $B$  is:
- $10\sqrt{2}$  km/h (North-East)
  - $10\sqrt{2}$  km/h (South-East)
  - $10\sqrt{2}$  km/h (North-West)
  - $10\sqrt{2}$  km/h (South-West)
- Q14.** A boat which has a speed of 5 km/h in still water crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water is:
- 1 km/h
  - 2 km/h
  - 3 km/h
  - 4 km/h
- Q15.** For a person moving with velocity  $\vec{v}_m$  in rain of velocity  $\vec{v}_r$ , the apparent velocity  $\vec{v}_{r/m}$  is:
- $\vec{v}_m - \vec{v}_r$
  - $\vec{v}_r + \vec{v}_m$
  - $\vec{v}_m \times \vec{v}_r$
  - $\vec{v}_r - \vec{v}_m$
- Q16.** A person standing on an open truck moving horizontally with constant speed  $v$  experiences rain falling at angle  $\theta$  with the vertical. When the truck accelerates forward, the angle  $\theta$  with the vertical will:
- Increase
  - Decrease
  - Remain unchanged
  - Become zero
- Q17.** Two particles  $A$  and  $B$  move with constant velocities  $\vec{v}_1$  and  $\vec{v}_2$ . At initial moment  $t = 0$ , their position vectors are  $\vec{r}_1$  and  $\vec{r}_2$ . The condition for collision of  $A$  and  $B$  is:
- $\vec{r}_1 - \vec{r}_2 = \vec{v}_1 - \vec{v}_2$
  - $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$
  - $(\vec{r}_1 - \vec{r}_2) \cdot (\vec{v}_1 - \vec{v}_2) = 0$
  - $(\vec{r}_1 - \vec{r}_2) \times (\vec{v}_1 - \vec{v}_2) = 0$
- Q18.** Raindrops are falling vertically with a velocity 10 m/s. To a cyclist moving due north, rain appears to come at  $45^\circ$  to vertical. The velocity of the cyclist is:
- 5 m/s due south
  - $10\sqrt{2}$  m/s due north
  - 10 m/s due north
  - 10 m/s due south
- Q19.** A boat can travel with speed  $v_{b/r}$  in still water. If river flow is  $v_r$  ( $v_r > v_{b/r}$ ), to achieve minimum drift the angle  $\theta$  of steering with the upstream direction must satisfy:
- $\cos \theta = \frac{v_{b/r}}{v_r}$
  - $\tan \theta = \frac{v_{b/r}}{v_r}$
  - $\sin \theta = \frac{v_r}{v_{b/r}}$
  - $\sin \theta = \frac{v_{b/r}}{v_r}$
- Q20.** A flag is mounted on a car moving due north with speed 9 m/s. Wind blows due east with speed 12 m/s. The direction in which the flag flutters is:
- $\tan^{-1}(3/4)$  south of east (South-East quadrant)
  - $\tan^{-1}(4/3)$  north of east
  - $\tan^{-1}(3/4)$  north of west
  - Directly South

# ANSWER KEY & STEP-BY-STEP SOLUTIONS

SMAPhysics.com | Curated by Syed Mazhar Ali

## Quick Answer Key

| Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans |
|------|-----|------|-----|------|-----|------|-----|
| 1    | (b) | 6    | (c) | 11   | (d) | 16   | (a) |
| 2    | (c) | 7    | (d) | 12   | (a) | 17   | (b) |
| 3    | (d) | 8    | (a) | 13   | (b) | 18   | (c) |
| 4    | (a) | 9    | (b) | 14   | (c) | 19   | (d) |
| 5    | (b) | 10   | (c) | 15   | (d) | 20   | (a) |

## Detailed Solutions

### Sol 1. Correct Option: (b)

For zero drift, the swimmer heads upstream at angle  $\theta$  to the normal such that  $v \sin \theta = u \Rightarrow \sin \theta = u/v$ .

Net speed across the river:

$$v_y = v \cos \theta = \sqrt{v^2 - u^2}$$

Time taken:

$$t = \frac{d}{v_y} = \frac{d}{\sqrt{v^2 - u^2}}$$

### Sol 2. Correct Option: (c)

$$\vec{v}_r = -30\hat{j}, \vec{v}_m = 10\hat{i}.$$

Relative velocity of rain:

$$\vec{v}_{r/m} = \vec{v}_r - \vec{v}_m = -10\hat{i} - 30\hat{j}$$

The rain appears to come from  $(10\hat{i} + 30\hat{j})$ .

Angle with vertical:

$$\tan \theta = \frac{10}{30} = \frac{1}{3}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right) \text{ towards east}$$

### Sol 3. Correct Option: (d)

$$\bullet \text{ Crossing time: } t = \frac{d}{v_{b/r}} = \frac{1 \text{ km}}{4 \text{ km/h}} = 0.25 \text{ h.}$$

$\bullet$  Drift along river flow:

$$x = u \times t = 3 \text{ km/h} \times 0.25 \text{ h} = 0.75 \text{ km}$$

### Sol 4. Correct Option: (a)

$$\text{Time to cross is } t = \frac{d}{v \cos \theta}.$$

To minimize  $t$ ,  $\cos \theta$  must be maximum ( $\cos \theta = 1 \Rightarrow \theta = 0^\circ$ ).

Hence, the swimmer must head directly perpendicular to the bank.

### Sol 5. Correct Option: (b)

$$\text{Let } \vec{v}_r = v_x \hat{i} - v_y \hat{j}.$$

$\bullet$  At  $v_m = 4 \text{ km/h}$ :  $\vec{v}_{r/m} = (v_x - 4)\hat{i} - v_y \hat{j}$ . Since it appears vertical,  $v_x - 4 = 0 \Rightarrow v_x = 4 \text{ km/h}$ .

$\bullet$  At  $v_m = 8 \text{ km/h}$ :  $\vec{v}_{r/m} = (4 - 8)\hat{i} - v_y \hat{j} =$

$$-4\hat{i} - v_y \hat{j}.$$

$$\tan(45^\circ) = \frac{4}{v_y} = 1 \Rightarrow v_y = 4 \text{ km/h.}$$

True speed of rain:

$$v_r = \sqrt{v_x^2 + v_y^2} = \sqrt{4^2 + 4^2} = 4\sqrt{2} \text{ km/h}$$

### Sol 6. Correct Option: (c)

When  $v_{b/r} < v_r$ , the angle for minimum drift sat-

$$\text{isfies } \sin \theta = \frac{v_{b/r}}{v_r} = \frac{3}{5} \Rightarrow \theta = 37^\circ.$$

$$\cos \theta = \frac{4}{5}.$$

$$\bullet v_x = v_r - v_{b/r} \sin \theta = 5 - 3 \left(\frac{3}{5}\right) = \frac{16}{5} \text{ km/h.}$$

$$\bullet v_y = v_{b/r} \cos \theta = 3 \left(\frac{4}{5}\right) = \frac{12}{5} \text{ km/h.}$$

Minimum drift:

$$x_{\min} = d \left(\frac{v_x}{v_y}\right) = 400 \times \frac{4}{3} \approx 533.3 \text{ m}$$

### Sol 7. Correct Option: (d)

$$\text{When standing still: } \tan(30^\circ) = \frac{v_x}{v_y} \Rightarrow v_x =$$

$$v_y \tan(30^\circ) = \frac{v_y}{\sqrt{3}}.$$

When running at 10 km/h, rain appears vertical:

$$v_x - 10 = 0 \Rightarrow v_x = 10 \text{ km/h}$$

$$v_y = \sqrt{3}v_x = 10\sqrt{3} \text{ km/h}$$

Actual speed:

$$v_r = \sqrt{10^2 + (10\sqrt{3})^2} = 20 \text{ km/h}$$

### Sol 8. Correct Option: (a)

$$\sin \theta = \frac{u}{v} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta =$$

$30^\circ$  upstream of normal.

Angle with river flow:

$$\alpha = 90^\circ + 30^\circ = 120^\circ$$

### Sol 9. Correct Option: (b)

$$\bullet T_1 = \frac{x}{v+u} + \frac{x}{v-u} = \frac{2vx}{v^2 - u^2}.$$

$$\bullet T_2 = \frac{2x}{\sqrt{v^2 - u^2}}.$$

Ratio:

$$\frac{T_1}{T_2} = \frac{v}{\sqrt{v^2 - u^2}} = \frac{1}{\sqrt{1 - \frac{u^2}{v^2}}}$$

**Sol 10. Correct Option: (c)**

$$\vec{v}_r = (12\sqrt{3} \sin 30^\circ)\hat{i} - (12\sqrt{3} \cos 30^\circ)\hat{j} = 6\sqrt{3}\hat{i} - 18\hat{j}.$$

For rain to appear vertical to cyclist ( $\vec{v}_c = v_c\hat{i}$ ):

$$6\sqrt{3} - v_c = 0 \implies v_c = 6\sqrt{3} \text{ m/s}$$

**Sol 11. Correct Option: (d)**

For minimum time, the swimmer heads perpendicular to flow ( $\theta = 0^\circ$ ).

$$T_{\min} = \frac{d}{v}.$$

$$\text{Drift } x_1 = u \times T_{\min} = \frac{ud}{v}.$$

**Sol 12. Correct Option: (a)**

$$\vec{v}_m = 3\hat{j} \text{ (North)}, \vec{v}_{r/m} = -4\hat{k} \text{ (Downwards)}.$$

$$\vec{v}_r = \vec{v}_{r/m} + \vec{v}_m = 3\hat{j} - 4\hat{k}.$$

$$\text{Speed} = \sqrt{3^2 + 4^2} = 5 \text{ km/h}.$$

$$\text{Direction: } \tan \theta = \frac{3}{4} \implies \theta = \tan^{-1}(3/4) \text{ south of vertical (falling from north towards south).}$$

**Sol 13. Correct Option: (b)**

$$\vec{v}_A = 10\hat{i}, \vec{v}_B = 10\hat{j}.$$

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B = 10\hat{i} - 10\hat{j}.$$

$$\text{Magnitude: } 10\sqrt{2} \text{ km/h}.$$

$$\text{Direction: East } +x, \text{ South } -y \implies \text{South-East}.$$

**Sol 14. Correct Option: (c)**

$$\begin{aligned} \text{Shortest path} &\implies \text{zero drift} \implies v_y = \sqrt{v_b^2 - v_r^2}. \\ t &= 15 \text{ min} = 0.25 \text{ h}. \end{aligned}$$

$$v_y = \frac{d}{t} = \frac{1 \text{ km}}{0.25 \text{ h}} = 4 \text{ km/h}$$

$$\begin{aligned} 4 &= \sqrt{5^2 - v_r^2} \implies 16 = 25 - v_r^2 \\ v_r &= 3 \text{ km/h} \end{aligned}$$

**Sol 15. Correct Option: (d)**

By vector subtraction definition of relative motion:

$$\vec{v}_{r/m} = \vec{v}_r - \vec{v}_m$$

**Sol 16. Correct Option: (a)**

$$\tan \theta = \frac{v_{\text{truck}}}{v_{\text{rain}}}.$$

As the truck accelerates,  $v_{\text{truck}}$  increases, so  $\tan \theta$  increases  $\implies \theta$  increases.

**Sol 17. Correct Option: (b)**

$$\text{Position of } A \text{ at time } t: \vec{R}_A(t) = \vec{r}_1 + \vec{v}_1 t.$$

$$\text{Position of } B \text{ at time } t: \vec{R}_B(t) = \vec{r}_2 + \vec{v}_2 t.$$

$$\text{For collision: } \vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t \implies \vec{r}_1 - \vec{r}_2 = (\vec{v}_2 - \vec{v}_1)t.$$

Since  $t > 0$  is a positive scalar, the unit vectors must be equal:

$$\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$$

**Sol 18. Correct Option: (c)**

$$\vec{v}_r = -10\hat{j}. \text{ Cyclist moving north: } \vec{v}_c = v_c\hat{j}.$$

$$\vec{v}_{r/c} = -v_c\hat{j} - 10\hat{j} \text{ (or horizontally relative component).}$$

For cyclist moving in  $+y$  direction with rain falling vertically:  $\vec{v}_{r/c} = -v_c\hat{i} - 10\hat{j}$ .

$$\tan(45^\circ) = \frac{v_c}{10} = 1 \implies v_c = 10 \text{ m/s due north}.$$

**Sol 19. Correct Option: (d)**

$$\text{Drift } x = \frac{d}{v_{b/r} \cos \theta} (v_r - v_{b/r} \sin \theta).$$

$$\text{Setting } \frac{dx}{d\theta} = 0 \text{ yields:}$$

$$\sin \theta = \frac{v_{b/r}}{v_r}$$

**Sol 20. Correct Option: (a)**

The flag flutters in the direction of wind relative to the car:

$$\vec{v}_{w/c} = \vec{v}_w - \vec{v}_c = 12\hat{i} - 9\hat{j}$$

$$\text{Direction: } \tan \theta = \frac{9}{12} = \frac{3}{4} \text{ south of east}.$$

Copyright 2026 **SMAPhysics.com** | Designed & Curated by **Syed Mazhar Ali** | All Rights Reserved.